Understanding the Infra: Essential Networking & Security for DevOps

1. **Client-Server Architecture**:

```
+----------+        Request        +----------+
| Client  | --------------------> | Server  |
| (e.g. PC)|                 | (e.g. Web)|
|   💻   | <-------------------- |  🖥️   |
+----------+        Response       +----------+
(Asks for data)              (Provides data)
```

Concept: A client machine requests information or services from a server machine, which processes the request and sends back a response.

2. **HTTPS Secure Web Traffic**:

```
+----------+      +----------------------------+      +----------+
| Browser | <=====>|  HTTPS (TLS/SSL Tunnel 🔒) |<=====> | Web     |
|   🌐   |      |      (Port 443)       |      | Server 📄 |
+----------+      +----------------------------+      +----------+
```

Concept: Your browser communicates with a website through an encrypted tunnel (HTTPS using TLS/SSL) on port 443, protecting the data exchanged.

3. **SSH Secure Remote Access**:

```
+----------+      +----------------------------+      +----------+
| Your PC | <=====>|  SSH Secure Tunnel ( 🔒 )  |<=====> | Remote  |
|   💻   |      |      (Port 22)        |      | Server 🖥️ |
+----------+      +----------------------------+      +----------+
```

Concept: You connect to a remote server securely over an encrypted SSH tunnel, typically on port 22, allowing command-line access.

4. **Firewall Filtering Traffic**:

GOOD Traffic (e.g. Port 22/SSH)

Internet --->-------------------------------->--------> Server 🖥️

🌍 | +-----------------+ | (Allowed)

| | Firewall Rules | |

| | (Allow Port 22) | |

| | (Block Others) | |

| +-----------------+ |

Internet --->--X BAD Traffic (e.g. Other Ports) X--> (Blocked)

🌍

Concept: The firewall inspects traffic, allowing connections on permitted ports (like SSH on 22) while blocking others based on its rule set.

5. **Authentication Methods (Conceptual):**

+-----------------------------+

| How Identity is Verified? |

+-----------------------------+

| ▢ Something You KNOW | -----> Password 🤫

| ▢ Something You HAVE | -----> Key 🔑 / Token 🎟️

| ▢ Something You ARE | -----> Biometric 👍 👀

+-----------------------------+

(Often combined for Multi-Factor Auth)

Concept: Authentication relies on proving who you are through knowledge, possession, or inherent traits.

6. **SSH Key Pair Authentication**:

Attempting SSH Login from Your PC...

| Uses Private Key 🔑

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+-------------------------+ +-----------------+

| Remote Server | | Your PC |

|-------------------------| |-----------------|

| Checks `authorized_keys`| | |

| Finds matching | ---- | Holds unique |

| Public Key (Lock) 🔓 | <--- | Private Key 🔑 |

+-------------------------+ +-----------------+

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✅ Access Granted (Keys Match!)

Concept: The server uses the public key (lock) to verify the authenticity of the connection attempt made using the corresponding private key (the only key that fits).

7. **SSH Access Setup Flow (Sequence):**

[Step 1: Generate Keys ✨ (🔑 + 🔓 on Your PC)] -->

[Step 2: Configure Server Firewall 🛡️ (Allow Port 22)] -->

[Step 3: Upload Public Key ⬆️ 🔓 (Add 🔓 to Server)] -->

[Step 4: Ensure Server is Ready ☁️ (SSH Service Running)] -->

[Step 5: Connect 🚀 (Use 🔑 from Your PC)]

Concept: A sequential process to set up and use secure key-based SSH access.